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Best Bike Paths, Luca Suzani, Alberto
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Requirement Analysis and Specification Document

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Chapter 1

Introduction

1.1 Purpose

The purpose of the *Best Bike Paths (BBP)* system is to promote cycling as a sustainable, safe, and conscious means of transportation by providing cyclists with a digital platform for collecting, accessing, and sharing information about cycling routes and activities.
The system addresses the need for reliable and up-to-date data on bike paths, allowing users to track their performance while contributing to the improvement of shared information.
Through active community participation, BBP aims to identify and maintain the safest and most efficient routes, fostering a more informed and widespread cycling mobility.
Ultimately, BBP seeks to enhance the overall cycling experience by integrating multiple functionalities and creating a collaborative, continuously updated information network driven by user contributions.

1.1.1 Goals

G1. **Support cyclists in tracking their activities:** Enable users to record their cycling routes and access detailed statistics such as total distance, average speed, and other performance indicators, enriched, when available, by meteorological data from external services (weather conditions, temperature and wind speed).

G2. **Allow cyclists to contribute information about bike paths:** Provide two modes for entering information related to bike paths:

- **Manual mode:** the user manually inputs the path data and its condition.
- **Automatic mode:** the system automatically collects data via GPS, accelerometer, and gyroscope during the cycling activity, requiring user confirmation before publication.

G3. **Enable the publication of recorded path information:** Allow users to make their recorded path information public, contributing to a shared and valuable database for the entire cycling community.

G4. **Enable route search and visualization:** Allow users to search for and display on a map one or more cycling routes connecting a chosen origin and destination, ranking them based on overall condition and effectiveness in reaching the goal.

G5. **Maintain up-to-date and reliable path information:** Ensure continuous updating of path data through the aggregation and analysis of user feedback, taking into account data freshness, consistency among reports, and user confirmations.

1.2 Scope

The main goal of the *Best Bike Paths (BBP)* system is to provide cyclists with a mobile application that allows them to record their activities and share information about bike routes, while also fostering the creation of an active and collaborative community. The application enables registered users to save their rides, view detailed statistics such as total distance covered or average speed, and contribute to updating the path inventory by reporting the condition of routes or the presence of potential obstacles. The information entered by users can be made public and is then aggregated by the system, which combines data from multiple sources to offer a comprehensive and up-to-date representation of the paths' conditions. Unregistered users can still take advantage of the consultation features by searching for routes between two points and viewing the available options on a map, ranked according to a score that considers both the quality and effectiveness of the routes. The system also interacts with an external meteorological service, which provides weather data to enrich the information related to recorded activities.

1.2.1 World Phenomena

1.2.2 Shared Phenomena

1.3 Definitions, Acronyms, and Abbreviations

1.3.1 Acronyms

- **BBP:** Best Bike Paths
- **DB:** DataBase

1.4 Revision History

1.5 Reference Documents

1.6 Document Structure

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Chapter 2

Overall Description

2.1 Product Overview

2.1.1 Scenarios

A user creates a path manually

1. The user logs into the platform using their credentials.
2. Once authenticated, the user accesses the "My Paths" section of their profile.
3. The user selects the "Add Path" option to start creating a new route.
4. The system opens a dedicated page for creating the path.
5. The user enters the required information, including:
    - the name of the path,
    - the streets that compose the path,
    - the condition of each segment,
    - any obstacles or issues present along the route.
6. After completing the input, the user confirms the creation of the path.
7. The system saves the data and displays a confirmation message.
8. The system also offers the user the option to make the path public, so it can be shared with the community.
9. The newly created path is displayed in the "My Paths" section of the user's profile.

A user searches for a path

1. The user can choose to:
    - log in with their account, or
    - use the platform without authentication.
2. From the home page, the user enters the starting point and the desired destination in the search bar.
3. The system receives the request and queries its database of published paths.
4. The system processes the available options and calculates a score for each path based on:

- the condition of the segments,
- the freshness of the information,
- the effectiveness of the route in connecting the selected points.

5. The system displays the results on the map, ordered in descending order of score.
6. The user reviews the results and can select a specific path to view more details.
7. The system displays detailed information for the selected path, such as:
    - total distance,
    - estimated travel time,
    - overall condition of the route.

User publishes a path

1. The user accesses the BBP system and opens the “My Routes” section.
2. The system displays the list of paths associated with the user, indicating for each one, among other properties, its status (private or public).
3. The user selects a private route and, using the appropriate button, requests its publication.
4. Upon receiving the publication request, the system asks the user to confirm additional information related to the route:
    - In the case of manually created routes, the user can review and, if necessary, modify the additional information previously entered during creation.
    - In the case of automatically generated routes, the user must confirm or correct the acquired information (if not already done) and may add further details manually.
5. Once the review is complete, the user confirms the publication of the route.
6. The system verifies the data consistency, assigns a unique identifier to the route, and makes it available to the community within the search and browsing functionality.
7. The system notifies the user of the successful publication and updates the route status from private to public.

User modifies the additional information related to a route

1. The user accesses the Best Bike Paths (BBP) platform and opens the “My Routes” section.
2. The system displays the list of routes associated with the user, indicating for each one, among other properties, its status (private or public).
3. The user selects a route and opens the section dedicated to additional information.
4. The system displays the current information, including:
    - The condition of the roads that make up the route;
    - The presence of obstacles or damaged sections;
    - The origin of each piece of data (manually entered or automatically acquired).
5. The user can modify, correct, or add new information:

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- If the route was created manually, the user can freely update existing data.
- If the route was created automatically, the user can review the information acquired by the system and/or add new data manually; in any case, they must confirm or correct the automatically acquired information before publication.

6. The system saves the changes and updates the database.

2.2 Product Functions

Sign-Up and Login

Sign-Up

The system lets the users register in order to manage and record data associated with each of them. When the user accesses the sign-up page for the first time, he's offered the option to log-in with one of their existing accounts through an OAuth service. This way, his BBP account is linked to another account that he owns, making it easier to remember and access for him. The system also gives the user the possibility of entering his email and a new password and to create an account directly inside BBP. In that case, BBP checks that the password is complex enough and that no other account is associated with the same email. If the checks pass, the system stores the information in the DB and grants the user access to the platform.

Login

If the user already has a BBP account or has previously linked their BBP account to one of the other existing accounts that he has through OAuth, the system grants them access the page. If the user access the page through OAuth, a button redirects them to the external account provider's website, which handles the login process and sends the user back only when he completes the login procedure. If, on the other hand, the user registered directly on BBP, the system first checks whether an account associated with the provided email, and then verifies that the password matches. If thats the case the user is redirected to the website's homepage.

Accessing BBP without accounts

The user can also access BBP without logging in. In this case, the functionalities that are available are more limited. The system lets the user browse between public paths, but all the recording features and the private paths needs the login to work

Activity Recording

The system allows registered users to record their cycling activities. When the user starts the recording, BBP continuously acquires real-time data from the mobile device, including:

- GPS coordinates used to compute distance, average speed, and elevation;
- timestamps associated with all collected events;
- accelerometer and gyroscope data to detect irregularities on the road surface;
- potential stops inferred from speed variations.

During or after the trip, the system processes the collected data and generates a summary reporting distance, duration, average speed, and detected anomalies. If the user decides to save the activity, BBP stores it in the database and makes it available for personal statistics.

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Automatic Path Creation

When the user enables the automatic mode, BBP reconstructs the bike path followed during the activity. This includes:

- continuous acquisition of GPS coordinates;
- segmentation of the route based on street structure and direction changes;
- detection of potholes, vibrations, or obstacles through accelerometer and gyroscope data;
- estimation of the user's speed to determine whether they are biking.

At the end of the activity, BBP generates a draft path containing:

- a map of the complete route;
- a list of detected street segments;
- automatically identified road conditions and anomalies.

Since the automatically detected data may contain false positives, the user must review, confirm, or correct all generated information before it becomes publishable. After confirmation, the path and related annotations are stored and made available to the community.

Meteorological Data Integration

When available, BBP enriches recorded activities with meteorological information obtained from an external weather service. After an activity is completed, the system sends:

- the activity timestamps;
- the geographical coordinates of the route.

The external service returns weather conditions for the corresponding time and location, including:

- temperature;
- general weather conditions (e.g., sunny, rainy, cloudy);
- wind speed and direction;
- possible weather alerts.

BBP associates the retrieved meteorological data with the recorded activity before storing it. These details are shown in the activity summary and are available for statistical analysis.

Manual Path Creation

In addition to the automatic creation of bike paths, registered users can also create paths manually by directly entering the required data. During this process, the platform asks the user to specify the name of the path, the streets that compose it, the condition of each segment, and the possible presence of obstacles or issues. Once the information has been confirmed, the system stores the data in its database and allows the user to decide whether to make the path publishable, meaning visible to the entire community. If the path is not made publishable, it remains visible only to the user who created it, within their personal profile.

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Path Search by Origin and Destination

Any user, including unregistered ones, can search for a bike path between two points using the search algorithm provided by the platform. The system queries its database of publishable paths and identifies those that connect the specified origin and destination points. The results are displayed on an interactive map and sorted in descending order based on a path score automatically computed by the platform. The user can select one of the displayed paths to view detailed information, such as the total distance, estimated travel time, and overall path conditions.

Path Ranking

The score assigned to each path is determined by considering several factors:

- Path condition, where better-maintained paths receive higher scores;
- Freshness of information, favoring paths with more recent data;
- Path effectiveness, meaning how efficiently the path connects the two selected points.

Entering route condition information

Each route, whether derived from an automatically recorded activity or manually created by the user, is characterized by a set of additional information describing its condition and rideability. In particular, this information concerns:

- The condition of the roads that make up the route;
- The presence of obstacles, such as potholes, surface damage, or dangerous sections.

Additional information is always managed through the same interface, but with different behavior depending on how the route was created:

- **Manual mode** During route creation, the user directly enters information about the condition of the roads and any obstacles.
- **Automatic mode** The system acquires data via the device's sensors (GPS, accelerometer, gyroscope); therefore, the user finds this information already filled in. Each piece of information must be confirmed.

Regardless of the mode, the user can always modify or update the additional information before and after publication, thus contributing to improving the quality of shared data.

Paths publication

Registered users can choose to publish a route, making it visible and accessible to other users within the platform. In particular, publication adds the route to *BBP*'s public archive, making it available to all other users, including non-registered ones, through the route search and browsing functionality. Only published routes can be viewed by others and contribute to enhancing the quality of shared community data. Publication always concerns the complete route, including all additional information such as road conditions and the presence of any obstacles.

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Merging information from different users

When multiple users provide information regarding the same route or overlapping sections, the BBP system automatically merges the received data concerning road conditions and the presence of obstacles. The merging process takes into account two main factors:

- The freshness of the information, meaning how recently it was provided;
- The number of confirmations received from different users for the same condition or status.

In the case of conflicting reports, the system applies a weighted evaluation mechanism that prioritizes the most recent and most frequently confirmed information.

2.3 User Characteristics

2.3.1 Registered User

A registered user has a personal profile containing the information provided during registration, the saved paths, and the cycling activities they have performed. They can save their rides, view detailed statistics (such as distance, average speed, and total time), and contribute to updating the path inventory by reporting the condition of the routes or the presence of obstacles. The registered user can also create new paths, either manually by entering data directly or automatically during an activity, using the sensors of their mobile device. After creation, the user can make these paths publishable, allowing the community to access and view them. Finally, the user can search for bike paths between two points and view them on the map, ordered according to the score assigned by the system.

2.3.2 Unregistered User

An unregistered user can access a limited version of the application, which only allows the consultation of publishable paths available on the platform. They can search for paths between two points, display them on the map, and view basic information such as the name, score, and general condition of each path, but they cannot perform any additional actions on them.

2.4 Assumptions, dependencies and constraints

2.4.1 Assumptions

2.4.2 Dependencies

D1. **External weather service** The system relies on the availability and accuracy of GPS data from users' devices to record cycling activities and create paths.

D2. **Mapping service** Offers geographic data and map layers used to visualize routes, match GPS tracks to road segments and retrieve details about paths, intersections, and terrains.

D3. **Storage database** Stores all platform data, including user routes, additional information (road conditions, obstacles), user profiles and publication metadata. Ensures persistence, consistency, and efficient retrieval.

D4. **Authentication services** Manages user identity verification, login processes and secure access to personal data and features.

D5. **Icon service** Provides a library of graphical icons used across the interface for consistent and intuitive representation of road conditions, obstacles, route statuses, and other UI elements.

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2.4.3 Constraints

C1. **Privacy regulations** The system must comply with relevant data protection laws and regulations (e.g., GDPR) to ensure user data privacy and security.

C2. **Automatic recording accuracy limits** The automatic activity recording feature will operate within the technical accuracy limits of mobile device sensors (GPS, accelerometer, gyroscope). The system must account for possible measurement noise, positioning errors and sensor variability across devices.

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Chapter 3

Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

Below are the main screenshots of the application. The first ones show the features available to both logged-in users and anonymous visitors, such as the homepage and the public path search.

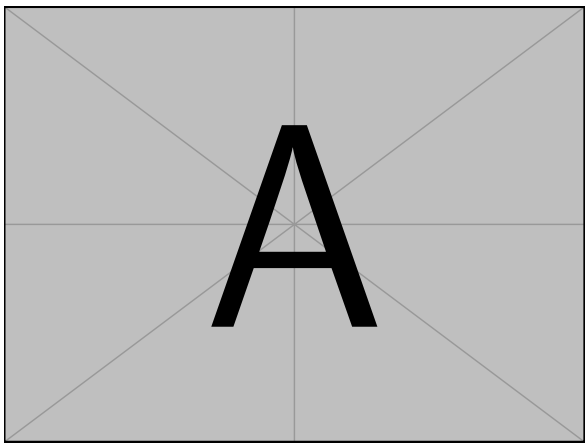


Figure 3.1: Homepage view

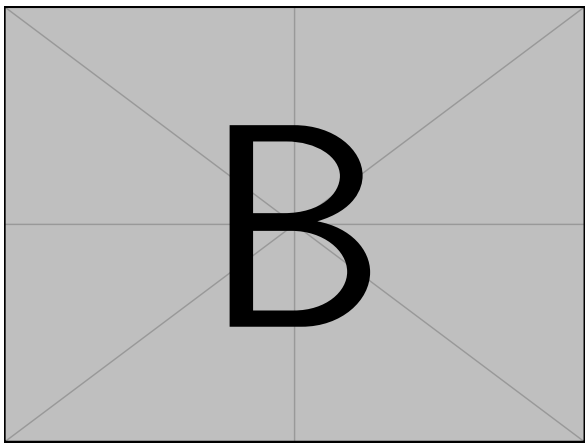


Figure 3.2: Public path search

Figure 3.3: Publicly accessible interfaces

The following screens are accessible only after authentication: the profile page, the activity recording interface, and the section where users can view their personal paths, both public and private.

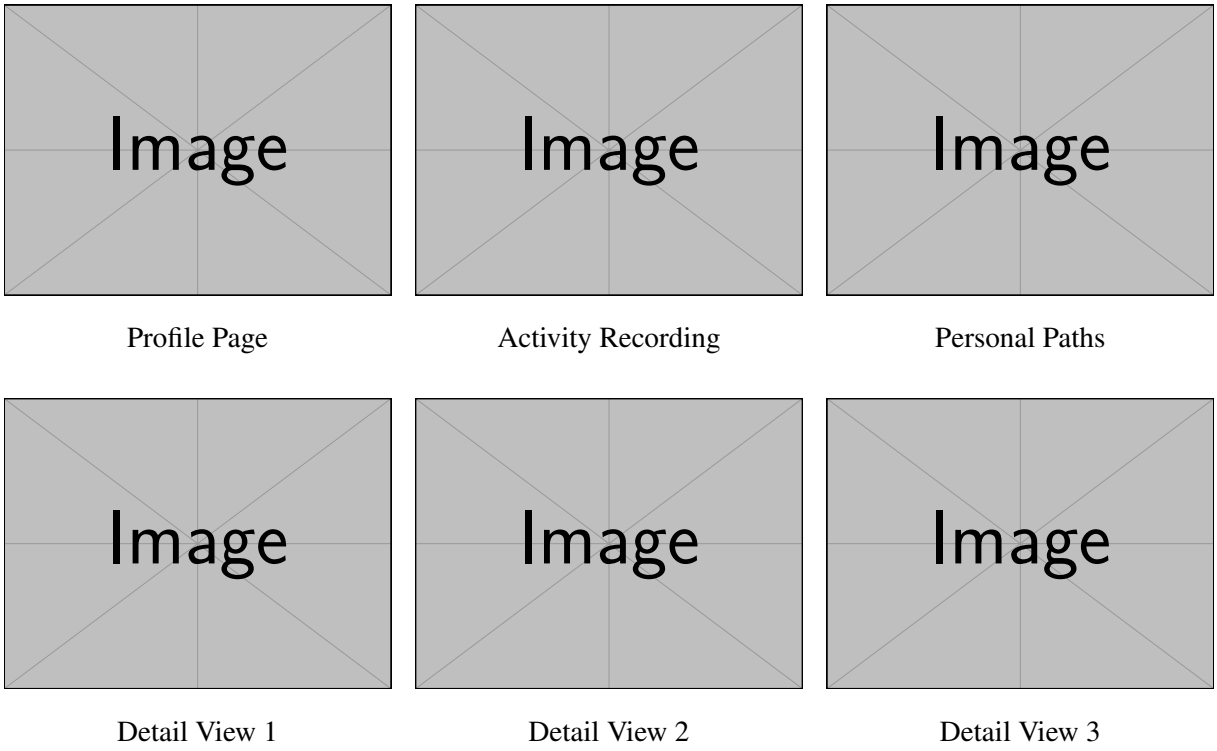


Figure 3.4: Authenticated User Interfaces

3.1.2 Hardware Interfaces

Since this is a web application, no additional hardware is required beyond a device capable of connecting to the internet. This applies to path browsing and manual activity creation.

However, to use the automatic recording feature, the device must include specific sensors—namely an accelerometer, GPS, and a compass.

3.1.3 Software Interfaces

- **Database:** To store user credentials, paths, and activities, the system must rely on a dedicated database accessible by the platform.
- **Identity provider:** The system may allow users to register with an email and password or authenticate through an external OAuth provider. This lets users access the platform without creating new credentials, using an account they already have with another service. In this case, an interface to communicate with the chosen provider is required to validate the user's identity.
- **Meteorological data provider:** When an automatic activity is recorded, the user can enrich it with weather information retrieved from an external API. To enable this, the system must expose a secure endpoint capable of requesting and receiving accurate meteorological data.

3.1.4 Communication Interfaces

The Best Bike Paths (BBP) system is a distributed application primarily relying on a Serverless architecture (e.g., using Function as a Service, FaaS) to manage backend processing. Communication interfaces must securely handle the transmission of sensitive data (such as user location) and high volumes of telemetric data.

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- **User and Client-Function Communication:** All interactions between clients (web browser interfaces and mobile devices) and the backend compute layer (serverless functions) must utilize HTTPS to ensure end-to-end encryption and data security.
- **Mobile Sensor Data Transmission:** The mobile application is responsible for securely transmitting data streams from device sensors (GPS, accelerometer, and gyroscope) directly to the database. These inputs are critical for the automatic trip tracking mode and the detection of road defects (potholes). The transmission mechanism must be optimized to minimize bandwidth consumption and battery drain, utilizing efficient, asynchronous upload of collected data.
- **External Interfaces (Meteorological Service):** The BBP system interacts with an external third-party meteorological service via its public APIs (a Machine-to-Machine interface). This external interface is used by the backend functions to retrieve real-time weather data (conditions, temperature, wind speed) to enrich the saved trip data.
- **Authentication and Authorization:** User access is managed via secure mechanisms (e.g., using OAuth). These mechanisms facilitate access control to the serverless functions that manage data storage and processing, ensuring only authenticated users can access their personalized features.

3.2 Functional Requirements

Sign Up & Login

The platform allows users to create an account, access the system, and recover credentials. Guest access is limited.

- **R1** The platform allows users to create an account by providing the required personal information.
- **R2** The platform allows registered users to log in and access the system.
- **R3** The platform allows registered users to recover the access credentials.
- **R4** The platform allows guest users (non-registered users) to access only the path search functionality of the system.

Activity recording

The system enables users to track their cycling activities and collect sensor data.

- **R5** The system shall allow registered users to start recording a new cycling activity.
- **R6** The system shall allow users to select an existing path from the path public repository to follow during the activity (not mandatory).
- **R7** The system shall collect raw data from the mobile device's sensors (GPS, accelerometer, gyroscope) during the recording of an activity.
- **R8** The system shall analyze the user's speed derived from GPS data to verify if the user is effectively cycling.
- **R9** The system shall analyze accelerometer and gyroscope data to detect potential obstacles or road surface anomalies.
- **R10** The system shall calculate real-time statistics during the recording, including total distance covered and average speed.
- **R11** The system shall allow the user to pause and resume the recording of an activity.

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Path creation

The system handles the creation of new paths through two modes: automatic detection from an activity or manual creation via map.

Automatic Creation

- **R12** Upon stopping an activity, the system shall display the list of automatically detected obstacles.
- **R13** The system shall require the user to validate the detected obstacles before proceeding.
- **R14** After validation, the system shall analyze the recorded GPS trace to identify if it matches an existing path in the database.
- **R15** If the recorded trace matches an existing public path, the system shall automatically associate the activity with that path entity.
- **R16** The system shall use the validated data to update the status and information of the existing public path.
- **R17** The system shall add the identified path to the “My paths” section if not already present.
- **R18** If the recorded trace doesn’t match any existing public path, the system shall allow users to generate a new path entity starting from the GPS trace of a recorded activity.
- **R19** The system shall allow the user to associate a name to the recorded path and store it in the “My paths” section.

Manual Creation

- **R20** The system shall allow the user to create a new path manually by selecting a sequence of streets from a map.
- **R21** The system shall require the user to manually input the status of the streets composing the path and mark the position and type of obstacles along it.
- **R22** The system shall allow the user to assign a name to the manually created path and store it in the “My Paths” section.

Path search from origin to destination

- **R23** The platform allows users to search for cycling paths based on a starting point and an ending point.
- **R24** The platform allows users to filter the search results based on the path parameters.

Path ranking

- **R25** The platform is able to rank the available paths based on the path data, including the condition of each segment, the freshness of the information, and the effectiveness of the path in connecting the selected points.

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Entering path condition information

- **R26** The platform is able to detect when a registered user follows a path that is already available to the public and associate the activity with that path.
- **R27** The platform allows registered users to update the condition information of a path, even without performing an activity on that path.

Path management and publication

- **R28** The system shall allow users to edit the information (name, street status, obstacles) of any path they have created.
- **R29** The system shall allow users to publish a stored path, making it available to the community.

3.2.1 Use case diagrams

3.2.2 Use cases

[UC1] User Registration

Name	**[Inserisci Nome Use Case]**
Actors	[Inserisci lista Attori]
Entry Condition	[Inserisci Condizione di Ingresso]
Event Flow	(a) [Primo passo del flusso...] (b) [Secondo passo del flusso...] (c) [Terzo passo del flusso...] (d) [Quarto passo del flusso...]
Exit Condition	[Inserisci Condizione di Uscita]
Exception	[Inserisci eventuali Eccezioni]

Table 3.2: Use Case: [Nome Use Case]

3.2.3 Sequence diagrams

3.2.4 Requirement Mapping

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Chapter 4

Formal Analysis Using Alloy

Organize this section according to the rules defined in the project description.

Chapter 5

Effort Spent

Chapter	Date	Alessio Pesavento	Alberto Spezzapria	Luca Suzani
Introduction	27/10/2025	1h	1h	1h
	07/11/2025	2h	2h	2h
	08/11/2025	-	1h	-
	09/11/2025	1h	-	-
	10/11/2025	-	-	1h
	Total	**4h**	**4h**	**4h**
Overall Description	11/11/2025	1.5h	1.5h	1.5h
	12/11/2025	-	1h	-
	13/11/2025	1.5h	1h	2h
	15/11/2025	1h	-	-
	29/11/2025	-	-	1h
	Total	**4h**	**3.5h**	**4.5h**
Product Functions	22/11/2025	2h	2h	2h
	25/11/2025	1h	1h	1h
	Total	**3h**	**3h**	**3h**

Table 5.1: Hours of work spent by each author on the *Introduction* and *Overall Description* chapters of the RASD document